

Application note: Flash Counter System

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Abstract

A crypto-box needs a flash counter system as part of its code. The subsystem makes use of a phototransistor to receive input. The design includes the 14M6245 phototransistor and the LM324 comparator to invert the input. The LTSpice simulation indicates that when there is no current in the circuit 14M6245 produces a high, and a low when there is no current. The implementation reflects the simulation results and proved that it was accurate. The FC subsystem is able to detect and display the flash counts.

Index Terms

comparator, 14M6245 phototransistor, LM325 comparator, LTSpice simulation, FC system

I. INTRODUCTION

This application note demonstrates a flash counter subsystem, or FC system that utilises a 14M6245 [1] phototransistor to detect light. This subsystem is part of a crypto-box system, which is a locked box that requires the correct code to open. The objectives of this application note is to design, simulate and implement the phototransistor that allows the subsystem to efficiently use the 14M6245 to detect flashes. Expected obstructions will be finding the appropriate components for the amplifier, inaccurate measurements due to human error or defected machinery and time pressure to deliver results. The expected results is for the voltage to change under different light conditions, the system to differentiate different light levels and for output to be detected by seven segment display.

The block diagram in figure 1 details the FC system. The 14M6245 detects the flash and is amplified by a comparator. The flash count is displayed on the seven segment display. The number on the display is checked and the result is visualised through a green LED, for correct, or a red LED, for incorrect. The green LED concludes the subsystem, and subsequently moves onto the next one. This application note will focus on the amplifier system of the FC subsystem.

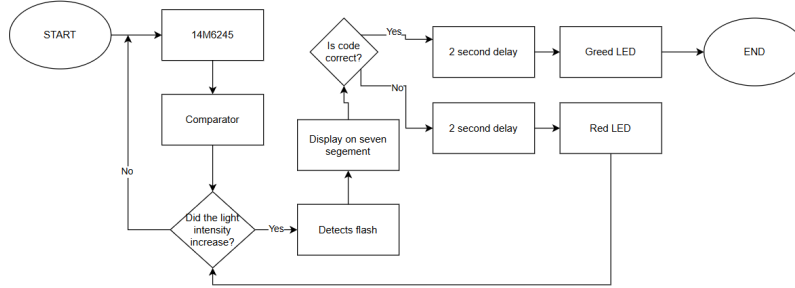


Fig. 1: System block diagram showing flash counter logic.

II. SYSTEM DESIGN

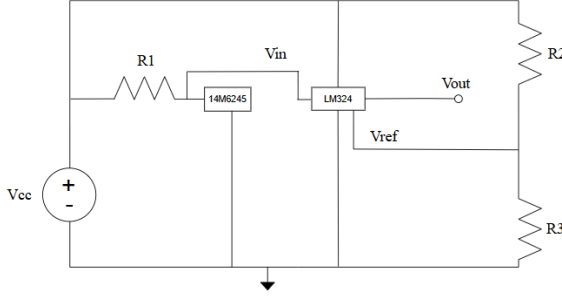
An amplifier is needed to increase the signal output from a sensor. Figure 2 shows the schematics for the circuit and simulation designs. Figure 2b is the system schematic for the simulation in LTSpice and will be discussed in the Simulation section. Figure 2a displays the circuit with the 14M6245 and comparator LM324 [2]. The circuit will receive a 5 V supply from a power supply. 14M6245 receives power through the collector after it passes through a 10 kΩ pull up resistor. LM324 is powered by the rail. The node between the collector and resistor is the inverted input of the LM324 and V_{ref} is supplied by the voltage divider, where R_2 and R_3 are 1 kΩ. From the equation

$$V_{ref} = \left(\frac{R_2}{R_3 + R_2} \right) V_{in} \quad (1)$$

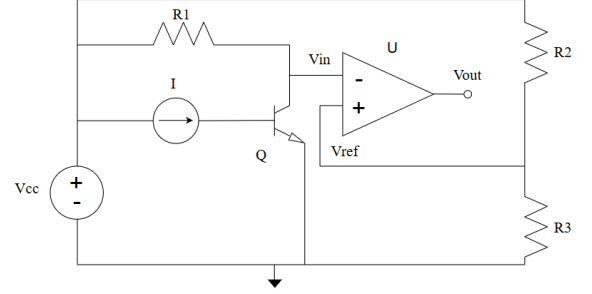
the value $V_{ref} = 2.5$ V is obtained. This value is used as the LM324 non-inverting input. Since $V_{ref} < V_{in}$, the comparator should produce a high when V_{in} is low, which is the signal needed by the seven segment display. The LM324 is the ideal comparator as it can amplify from 3 V – 36 V.

Figure 3 shows the characteristic plot of the 14M6245 phototransistor. As the distance between the light and the phototransistor decreased, the voltage decreased. This signifies that as light intensity increases, the voltage decreases, as seen in 2. This indicates a dark sensor.

$$I \propto 1/d^2 \quad (2)$$



(a) Design with components



(b) Design for simulation

Fig. 2: System schematics of phototransistor

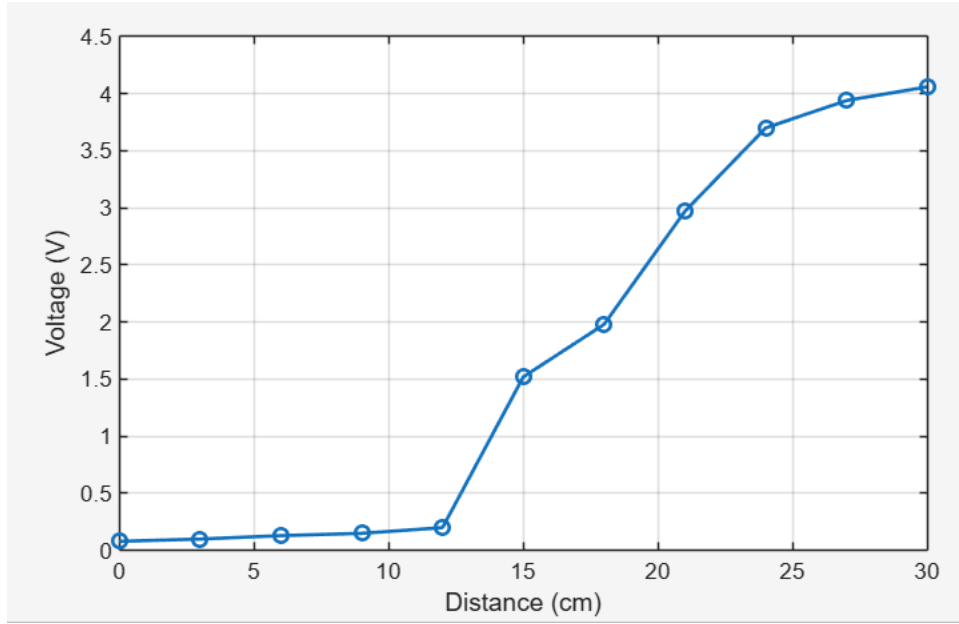


Fig. 3: Characteristic graph showing voltage versus distance of light

III. SIMULATION

A simulation is performed on the schematic shown in figure 2b using LTSpice. The 14M6245 and LM324 is replaced with a current source, an NPN transistor and universal op-amp. These components are replaced because the LTSpice component library does not contain the 14M6245 and LM324. The current source is set to pulse current to imitate the light flashing on and off. Therefore, it emulates a phototransistor to produce similar results. Measurements are taken from the transistor collector, labeled V_{in} , and the op-amp output, labeled V_{out} . In figure 4, when the current pulses on and off, V_{in} output is low then high, respective of those states. This relates to the characteristic graph in figure 3. As the light intensity increases, the voltage decreases, thus creating a dark sensor. V_{out} is the inverted output of V_{in} . When current is present, output is high, but low when current is absent.

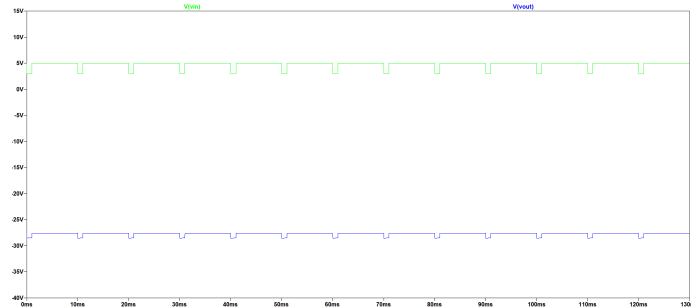


Fig. 4: LTSpice simulation showing inverted input and output of comparator

IV. IMPLEMENTATION

The circuit schematic from figure 2a is used to build the circuit on a breadboard. An oscilloscope is used to take measurements. Figure 5a is the collector output of the transistor, which is the inverted input of LM324. The output is high when the sensor is in the dark. The output is low when the light shines on the sensor. Figure 5b shows the inverted output of 5a which goes directly to the seven segment display and detects the flash. These measured results coincide with the simulated results on LTSpice, making the simulation accurate and confirms that this is a dark sensor. Figure 5a is not fully accurate as it contains noise that is caused by the bread board and the environment not being fully void of light, which interferes with the signal.

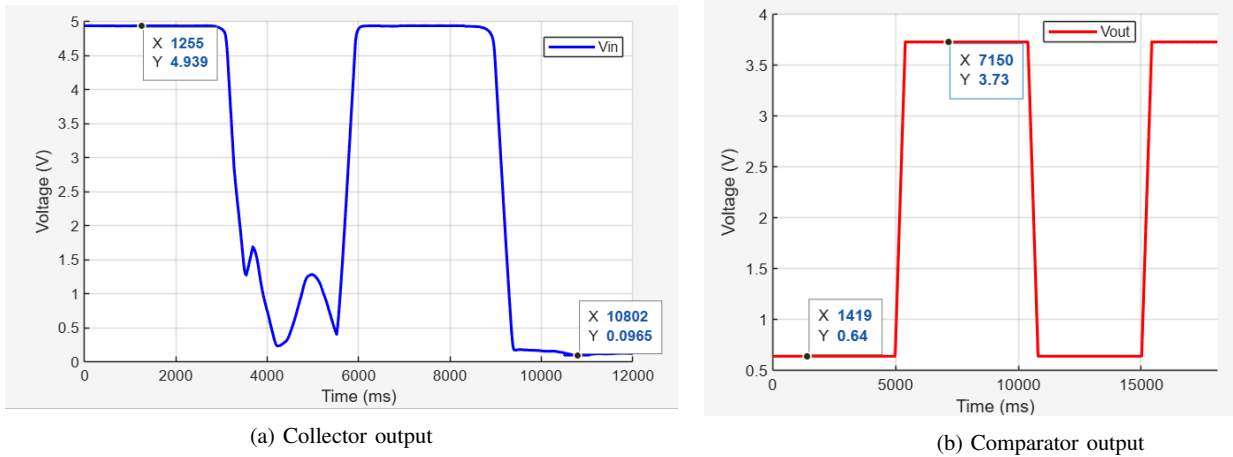


Fig. 5: Measured results

V. CONCLUSION

This application note showcased an FC system that detects and counts flashes. The design made use of the 14M6245 phototransistor and LM324 comparator to amplify the voltage and display the increased signal to the seven segment display. The simulations on LTSpice shows that the 14M6245 output is high when there is no current, and low when there is current. The LM324 converts this to low and high. In the implementation, these results verify the simulation results and thus, a working FC system was built.

REFERENCES

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- [2] *LMx24, LMx24x, LMx24xx, LM2902, LM2902x, LM2902xx, LM2902xxx Quadruple Operational Amplifiers*, Texas Instruments Incorporated, Dallas, TX, September 2025, data sheet. [Online]. Available: <https://www.ti.com/lit/ds/slos066ae/slos066ae.pdf?ts=1760899041789>